DATA SCIENCE CAPSTONE · FINAL PRESENTATION

OpenArt

Find the calls that fit your practice.

Recommender system and application assistant for artists and cultural workers

Luca Tomarelli

RESIDENCIES
GRANTS
COMMISSIONS

Artists lose real time searching and applying for calls buried in long eligibility requirements.

WHO IT IS FOR

Artists and cultural workers looking for residencies, grants and commissions across Europe.

RESEARCH QUESTION

Can NLP automatically identify eligibility requirements in open calls?

The data behind the model

479
CALLS COLLECTED

51
INSTITUTIONAL SOURCES

231
CALLS ANNOTATED

696
LABELLED SENTENCES

1
DISCOVER
Institutions' own public pages

2
EXTRACT
An LLM reads them into structured fields

3
TRANSLATE
French, Italian, German to English

4
ANNOTATE
By hand, into 9 requirement classes

5
MODEL
The graded core of the project

EVERY SENTENCE GETS ONE OF 9 LABELS

RESIDENCE | NATIONALITY | AGE | DISCIPLINE | CAREER STAGE | EDUCATION | STUDENT STATUS | OTHER ELIGIBILITY | NONE

You don't need a degree to make art, and where a call cares about career stage, it favours beginners.

NO DEGREE NEEDED

94%

of annotated calls don't require any formal education.

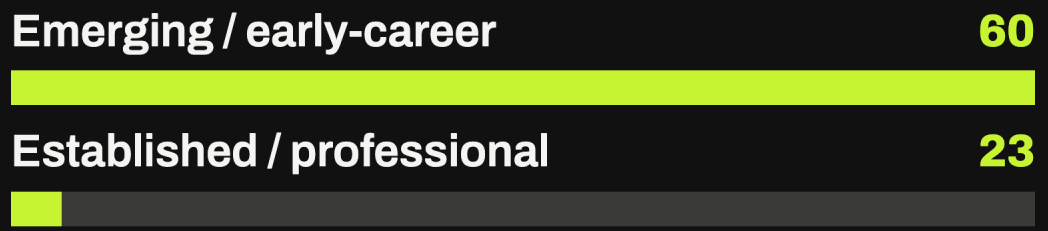
Based on 231 annotated calls.

WHEN CAREER STAGE MATTERS

76%

of calls set no career-stage bar at all.

OF THE 24% THAT DO



Funding and fees, across 479 calls

77%

of calls state some funding: a grant, stipend, materials, travel or housing.

Typical award: €15,000. That is the median; a few large institutional budgets skew the mean far higher.

3%

of all 479 calls charge any fee to apply.

Of the few that mention fees at all, more say no fee than yes.

Validation and training

VALIDATION

GROUPED BY CALL

Sentences from one call are near-identical, so they never straddle train and test.

5 FOLDS

With 696 sentences one split is noisy; five held-out folds are stable.

TRAINING

CLASS-WEIGHTED

Rare classes are not drowned out, and no real data is thrown away.

TUNED IN THE FOLD

Grid search runs on the training folds only; the held-out fold stays untouched.

Metrics and models

HEADLINE METRIC

Macro-F1

F1 blends precision and recall into one score per class. Macro averages it equally across all 9 classes, despite real imbalance.

Also tracked: accuracy, precision, recall, confusion matrix.

MODELS COMPARED, SAME FOLDS

1. **TF-IDF + Logistic Regression (baseline)**
2. **TF-IDF + Linear SVM**
3. **Embeddings (bge-large) + Logistic Regression**
4. **LLM few-shot (reference only)**

Representation: TF-IDF counts keywords, embeddings capture meaning. Classifier: models 1 and 2 share TF-IDF features and swap only the classifier, which isolates what the bottleneck was.

Answer to the research question: yes

74%

accuracy across 9 classes, on calls the model never saw.

> Multi-label upgrade: 76% of the labels in compound sentences are found, up from 37%.

Model	Macro-F1	Accuracy
TF-IDF + LogReg (baseline)	0.69 ± 0.02	0.69
TF-IDF + SVM	0.70 ± 0.02	0.68
bge-base + LogReg	0.75 ± 0.01	0.74
bge-large + LogReg (chosen)	0.76 ± 0.01	0.74
LLM few-shot (reference)	0.80	0.75

Macro-F1: mean ± std over 10 seeds. The LLM row is a single split.

The right trade-off

THE CHOSEN MODEL

0.76

macro-F1 · 0.74 accuracy

bge-large embeddings + Logistic Regression: the best-performing trained candidate.

WHY NOT THE LLM, WHICH SCORED HIGHER

$0

per call, running locally

An API cost would grow with every sentence and every artist. The ~0.04 F1 gap is the measured price of that choice.

CONFIDENCE THRESHOLD

67% auto-accepted

85% accurate on that set

The remaining 33% go to the artist for review.

The product, in one flow

1

DOCUMENTS

CV, statement and portfolio, uploaded once.

2

PROFILE

Your facts, read from the CV by rules.

3

PROJECTS

Your works, found in your documents.

4

DISCOVER

Calls ranked for you, each with a verdict.

5

DRAFT

An assistant helps you start your application.

Everything is built from your own documents.

Eligibility: from sentences to a verdict

1

SPLIT

The call text becomes sentences.

→

2

CLASSIFY

A trained model labels each sentence and how sure it is.

→

3

CHECK

Requirements are compared with your profile by rules.

→

4

VERDICT

Eligible, check or ruled out, with the sentence quoted.

When in doubt it says check, never a guessed no.

RAG: your documents, searchable

1
UPLOAD
CV, statement and portfolio.

→

2
SPLIT
Chunks that respect CV sections.

→

3
EMBED
Each chunk becomes a vector.

→

4
STORE
A private knowledge base per artist.

→

5
RETRIEVE
Closest passages, each with its source.

Retrieval feeds two things: your search query and the passages the assistant quotes.

Every passage keeps its source, like cv.pdf, page 1, Education.

Recommender: calls ranked for you

1

YOUR DOCUMENTS

Passages about your practice and what you look for become the query.

2

EVERY CALL SCORED

Each call gets a Match % from how close it is to your query.

3

THREE GROUPS

Fit, then another discipline, then not eligible. Sorted by Match %, nothing hidden.

Match % = how close the call is to your documents, plus the rare words you share, minus a discipline mismatch.

The assistant: grounded drafting

1

CHECKLIST

What to write and what to gather.

→

2

CONTEXT

The call, your profile, your projects and passages.

→

3

REVIEW

You edit and confirm every text.

→

4

DOWNLOAD

Unlocks when everything is reviewed.

An LLM drafts, you decide. It never invents a fact: a gap becomes a question or a placeholder.

OpenArt prepares the application. You submit it yourself.

OpenArt

THE ARTIST FOR THE DEMO

Elena Moretti

Interdisciplinary artist, based in Berlin

Born 1988, Buenos Aires. Works with moving image, installation, works on paper and collective pedagogy.

Film direction at ENERC. MA Art and Media, UdK Berlin.

Shown in Berlin, Buenos Aires and Barcelona, and at the Venice Biennale and HKW.

Portfolio and CV anonymized for the demo.



Surplus
2026 · 1-channel video, 14'30"



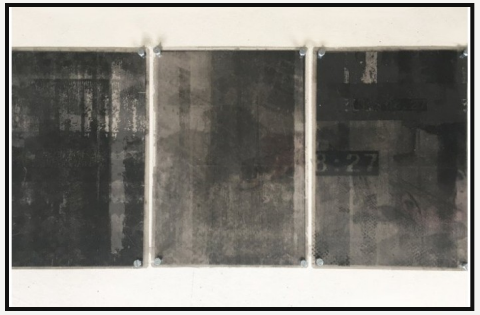
Memorias
2026–27 · feature film, in post-production



To Twist the Course of Signs
2025 · 3-channel video installation



City Loop
2018 · 6-channel video installation



Tríptico Difuso
2022 · ink on acetate, 60 × 40 cm each

7
works found automatically in her documents

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Thank you

Now, the live demo.

Luca Tomarelli

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